

ROMA CIAMPINO, Italy

WMO: 162390

Lat: 41.8081N

Lon: 12.5847E

Elev: 427

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.4	33.4	17.2	13.3	40.3	21.2	16.1	40.5	26.9	51.1	23.7	52.2	5.6	40	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.7	93.3	70.8	91.2	70.8	88.2	70.1	76.4	83.6	74.9	82.9	73.5	82.3	9.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	132.3	80.0	73.0	124.4	78.7	71.2	116.7	77.8	40.1	83.6	38.6	83.2	37.3	82.7	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	19.3	16.2	DB	26.4	97.7	2.7	3.0	24.5	99.9	22.9	101.6	21.4	103.2	19.4	105.4	(4)
(5)			WB	24.2	79.6	2.3	1.8	22.5	80.9	21.1	81.9	19.8	82.9	18.1	84.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	46.1	47.2	51.6	57.1	64.5	72.6	77.5	77.9	70.2	63.3	55.0	48.1
	DBStd	12.23	5.22	5.09	4.54	4.52	4.68	5.27	3.88	4.04	4.59	4.72	5.73	5.48
	HDD50	404	142	101	36	3	0	0	0	0	0	0	20	102
	HDD65	2731	587	497	416	240	67	4	0	0	9	86	301	524
	CDD50	4422	20	24	86	215	449	677	853	866	605	414	171	42
	CDD65	1275	0	0	0	3	51	232	388	401	164	34	2	0
	CDH74	11463	0	0	0	24	372	2080	3922	3925	1014	125	1	0
	CDH80	4413	0	0	0	2	60	725	1688	1709	224	5	0	0
Wind	WSAvg	6.3	6.3	6.8	6.9	6.7	6.2	6.2	6.1	5.7	5.8	5.6	6.7	6.3
Precipitation	PrecAvg	31.5	3.0	3.0	2.6	2.6	2.0	1.3	0.8	1.5	2.8	3.7	4.4	3.8
	PrecMax	45.9	8.3	7.8	5.6	8.3	5.3	5.0	4.3	5.1	6.5	8.7	9.6	10.2
	PrecMin	20.4	0.1	0.2	0.7	0.6	0.0	0.1	0.0	0.0	0.2	0.1	0.3	1.1
	PrecStd	6.8	2.2	2.0	1.4	2.0	1.2	1.2	1.1	1.4	1.9	2.7	2.4	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.5	62.9	69.9	77.0	85.8	93.2	95.4	97.2	89.3	80.5	71.8	64.2
		MCWB	54.0	52.1	54.8	61.0	65.8	70.7	71.5	71.0	68.7	66.0	62.4	57.9
	2%	DB	58.9	59.8	66.0	73.0	80.8	90.2	93.0	94.6	85.0	77.1	68.3	60.9
		MCWB	54.2	51.8	55.2	59.0	64.3	70.3	70.8	71.1	69.3	65.8	61.8	56.7
	5%	DB	57.0	57.6	62.9	69.7	78.4	87.5	91.0	91.3	82.4	75.0	66.4	59.0
		MCWB	52.6	50.9	53.7	58.0	63.4	69.5	70.5	71.3	67.9	65.4	60.6	55.0
	10%	DB	55.1	55.7	60.7	66.6	75.2	84.3	88.1	88.2	79.9	72.0	64.4	57.1
		MCWB	51.0	50.1	53.0	56.6	63.0	68.2	70.1	70.5	67.6	64.3	59.3	53.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.2	55.6	59.7	63.8	69.8	76.9	78.3	78.7	75.3	71.4	65.7	59.7
		MCDB	59.1	59.2	64.7	71.6	78.4	84.2	85.3	85.6	80.6	75.6	68.4	62.3
	2%	WB	55.0	54.0	57.2	61.4	67.5	74.1	76.1	76.8	73.0	69.0	63.9	57.5
		MCDB	58.0	57.5	62.3	69.0	76.2	83.4	83.8	83.6	79.0	73.2	66.8	60.1
	5%	WB	53.2	52.8	55.6	59.6	65.8	72.1	74.6	75.2	71.5	67.3	61.9	55.7
		MCDB	56.3	56.5	60.5	66.7	74.1	81.9	83.3	82.8	77.7	71.8	65.2	58.4
	10%	WB	51.6	51.3	54.1	58.1	64.3	70.4	73.2	73.7	70.0	65.6	59.9	53.8
		MCDB	54.5	55.0	59.3	64.7	72.3	80.2	82.5	82.4	76.7	70.4	63.6	56.8
Mean Daily Temperature Range	5% DB	MDBR	14.0	15.3	16.3	16.8	17.8	19.1	19.8	19.7	17.7	16.5	13.9	13.4
		MCDBR	13.9	16.7	19.7	20.8	22.1	22.2	22.6	23.3	20.2	18.6	14.5	12.6
		MCWBR	10.1	11.0	11.7	11.2	10.2	9.7	10.2	10.0	9.5	10.6	9.9	9.3
	5% WB	MCDBR	12.4	13.5	16.6	17.7	19.6	20.0	19.5	19.4	17.0	15.6	12.1	11.2
		MCWBR	10.1	10.3	11.2	11.1	10.9	10.5	10.7	10.9	10.2	10.8	9.7	9.3
Clear-Sky Solar Irradiance	taub	0.350	0.376	0.418	0.443	0.448	0.440	0.431	0.431	0.423	0.412	0.386	0.351	
	taud	2.399	2.301	2.180	2.134	2.169	2.244	2.274	2.285	2.299	2.292	2.295	2.390	
	Ebn at Noon	249	259	260	263	264	266	267	263	256	243	230	237	
	Edh at Noon	26	33	42	47	47	43	42	40	37	33	29	24	
All-Sky Solar Radiation	RadAvg	554	843	1236	1648	2015	2295	2351	2050	1495	1021	628	500	
	RadStd	55	84	142	115	169	99	108	112	74	88	59	54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.81	N/A	N/A	N/A	-0.90	-1.49	N/A	-139	N/A	N/A
Regional (18 neighbors)	+0.76	N/A	N/A	+0.95	N/A	N/A	N/A	-140	+243	+124

Nomenclature: See separate page